

PRACTICAL

TASK-1 :

1. Execution

```
bindhu@LAPTOP-NQPCRIKO: ~$ nano task1.c
bindhu@LAPTOP-NQPCRIKO: ~$ gcc task1.c
bindhu@LAPTOP-NQPCRIKO: ~$ ./a.out
Enter a Linux command: ls
Parent PID: 989
Child PID: 990
a.out  c2      dir6      'execex1.c*'  f3      newfile1  open2.cX
      pipe3.c  private.txt  script.txt
a7     dir1     dir9      exechole.c   fork.c   o          openb1.c
      pipe6.c  public.txt  student_portal
a8     dir2     exec1.c   f1           fork3.c  ope2.c    permission_
lab    pipe7.c   read1.c   task1
a9     dir3     exec2.c   f1.c         linux_TASK4  open1.c   pipe2
      pipeex.c  readwrite.c  task1.c
b1     dir5     exec3.c   f2           linux_task4  open2.c   pipe2.c
      pipeex1.c  readwrite.c.save  x1
Child process completed
bindhu@LAPTOP-NQPCRIKO: ~$ |
```

Task 2:

1. uname

```
bindhu@LAPTOP-NQPCRIKO: ~$ uname -a
Linux LAPTOP-NQPCRIKO 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
bindhu@LAPTOP-NQPCRIKO: ~$
```

2. lscpu

```
bindhu@LAPTOP-NQPCRIKO: ~$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:              48 bits physical, 48 bits virtual
  Byte Order:                 Little Endian
CPU(s):                       8
  On-line CPU(s) list:       0-7
Vendor ID:                    AuthenticAMD
  Model name:                 AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx
  CPU family:                 23
  Model:                     24
  Thread(s) per core:         2
  Core(s) per socket:         4
  Socket(s):                  1
  Stepping:                   1
  BogomIPS:                   4192.00
  Flags:                      fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc
                             p lm constant_tsc rep_good nopl xtopology tsc_r
                             eliable nonstop_tsc cpuid extd_apicid tsc_known
                             _freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_
                             2 movbe popcnt aes xsave avx f16c rdrand hyperv
                             isor lahf_lm cmp_legacy svm cr8_legacy abm sse4
                             a misalignsse 3dnowprefetch osvw topoext perfct
                             r_core ssbd ibpb vmcall fsgsbase bmi1 avx2 sme
                             p bmi2 rdseed adx smap clflushopt sha_ni xsaveo
                             pt xsavec xgetbv1 clzero xsaveerptr virt_ssbd a
                             rat npt nrip_save tsc_scale vmcb_clean flushbya
                             sid decodeassists pausefilter pfthreshold v_vms
                             ave_vmload

Virtualization features:
  Virtualization:             AMD-V
  Hypervisor vendor:          Microsoft
  Virtualization type:        full
Caches (sum of all):
  L1d:                         128 KiB (4 instances)
  L1i:                         256 KiB (4 instances)
  L2:                          2 MiB (4 instances)
  L3:                          4 MiB (1 instance)
NUMA:
```

```
bindhu@LAPTOP-NQPCRIKO: ~$ cat /sys/kernel/debug/dmidecode/dmidecode
rat npt nrip_save tsc_scale vmcb_clean flushbya
sid decodeassists pausefilter pfthreshold v_vms
ave_vmload

Virtualization features:
  Virtualization:      AMD-V
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 128 KiB (4 instances)
  L1i:                 256 KiB (4 instances)
  L2:                  2 MiB (4 instances)
  L3:                  4 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-7
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Reg file data sampling: Not affected
  Retbleed:             Mitigation; untrained return thunk; SMT vulnera
                        ble
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass:    Mitigation; Speculative Store Bypass disabled v
                        ia prctl
  Spectre v1:           Mitigation; usercopy/swapgs barriers and __user
                        pointer sanitization
  Spectre v2:           Mitigation; Retpolines; IBPB conditional; STIBP
                        disabled; RSB filling; PBRSE-eIBRS Not affecte
                        d; BHI Not affected
  Srbds:                Not affected
  Tsa:                  Not affected
  Tsx async abort:      Not affected
  Vmscape:              Not affected
bindhu@LAPTOP-NQPCRIKO: ~$
```

3. lsblk

```
bindhu@LAPTOP-NQPCRIKO: ~$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    1G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
```

4. ps

```
bindhu@LAPTOP-NQPCRIKO:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16    0 159.4M  1 disk
sdc    8:32    0    1G    0 disk [SWAP]
sdd    8:48    0    1T    0 disk /mnt/wslg/distro
/
```

5. top

```
bindhu@LAPTOP-NQPCRIKO:~$ top
top - 12:46:53 up 55 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  27 total,   1 running,  26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  2877.6 total,  2329.4 free,   458.4 used,   164.3 buff/cache
MiB Swap:  1024.0 total,  1024.0 free,    0.0 used.  2419.2 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
889 bindhu    20   0    8172    6116    4012 R   0.3   0.2   0:00.07
  1 root      20   0   24572   15460   11492 S   0.0   0.5   0:05.63
  2 root      20   0    3180    2204    2072 S   0.0   0.1   0:00.22
  6 root      20   0    3180    2044    1940 S   0.0   0.1   0:00.00
 61 root      19  -1   50396   16824   15528 S   0.0   0.6   0:01.69
 99 systemd+  20   0   22416   14404   11908 S   0.0   0.5   0:00.52
113 root      20   0   35244   12240    9256 S   0.0   0.4   0:02.49
134 root      20   0    2888    1952    1820 S   0.0   0.1   0:00.26
135 root      20   0    4472    3100    2856 S   0.0   0.1   0:00.05
136 message+  20   0    8828    5352    4668 S   0.0   0.2   0:00.29
142 root      20   0   44096   29704   14564 S   0.0   1.0   0:01.05
159 root      20   0   18452    9404    8148 S   0.0   0.3   0:00.40
190 syslog    20   0  220548    5780    4484 S   0.0   0.2   0:00.36
202 root      20   0    5492    2712    2528 S   0.0   0.1   0:00.04
238 root      20   0  123460   32572   17964 S   0.0   1.1   0:01.01
259 _chrony    20   0   23732   11080    7072 S   0.0   0.4   0:00.40
280 _chrony    20   0   12096    2404    1760 S   0.0   0.1   0:00.04
346 root      20   0    8648    5488    4644 S   0.0   0.2   0:00.15
426 bindhu    20   0   22332   13600   10988 S   0.0   0.5   0:00.52
430 bindhu    20   0   22916    4092    2088 S   0.0   0.1   0:00.00
460 bindhu    20   0    6136    5424    3864 S   0.0   0.2   0:00.12
701 root      20   0    3184    1116     980 S   0.0   0.0   0:00.00
702 root      20   0    3200    1256    1108 S   0.0   0.0   0:00.05
706 bindhu    20   0    6268    5612    3936 S   0.0   0.2   0:00.37
710 root      20   0    3184    1116     980 S   0.0   0.0   0:00.00
712 root      20   0    3200    1128     980 S   0.0   0.0   0:00.00
713 bindhu    20   0    6268    5540    3868 S   0.0   0.2   0:00.21
```